Unit 2 - Linear Algebra
Solving Systems of Equations

MSDS 520

Outline

Systems of Equations in Two Variables

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Systems of Equations in Two Variables

A (linear) equation in two variables might look like this:

$$ax + by = c$$

This cannot be solved for any particular values of x, y since there are two variables and only one equation.

A system of (linear) equations might look like this

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

This can be solved since there are both two variables and two equations.

Systems of Equations in Two Variables

Examples:

- Linear:

$$\begin{aligned} x + y &= 10 \\ x - 2y &= 3 \end{aligned}$$

- Nonlinear:

$$\begin{aligned} x^2 + y &= 10 \\ x - 2\sqrt{y} &= 3 \end{aligned}$$

Types of Linear Systems in Two Variables

Types of Linear Systems in Two Variables

Graphing the system of equations

$$a_1x + b_1y = c_1$$
$$a_2x + b_2y = c_2$$

would give two lines (one per equation). The lines can either:

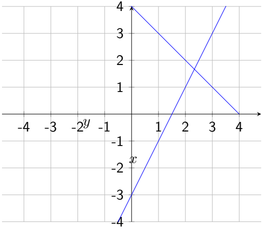
- Cross at one point (the one unique solution)
- Both lines overlap entirely (infinite solutions)
- They are parallel and don't cross (no solution)

Examples:

$$\begin{aligned} x + y &= 4 \\ 2x - y &= 3 \end{aligned} \qquad \begin{aligned} x + y &= 4 \\ 2x + 2y &= 8 \end{aligned} \qquad \begin{aligned} x + y &= 4 \\ x + y &= 3 \end{aligned}$$

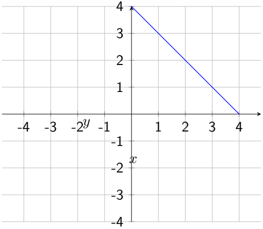
Linear Systems of Equations with Two Variables

$$x + y = 4$$
$$2x - y = 3$$



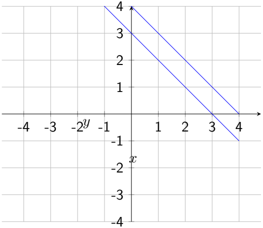
- One solution (lines cross at one point)
- Consistent
- Independent

$$x + y = 4$$
$$2x + 2y = 8$$



- Infinite solutions (lines overlap)
- Consistent
- Dependent

$$x + y = 4$$
$$x + y = 3$$



- No solutions (lines are parallel)
- Inconsistent

The Method of Substitution

The Method of Substitution

Solving a system with the method of substitution:

- Choose one variable, say x, and one equation
- Solve that equation for that variable
- Substitute the expression from the above step into the other equation for the other variable, say y (obtaining a value for it)
- Plug this value for y back into the first equation to obtain a value for x

Examples:

Consistent Independent	Consistent Dependent	Inconsistent
$$\begin{aligned} x + y &= 4 \\ 2x - y &= 3 \end{aligned}$$	$$\begin{aligned} x + y &= 4 \\ 2x + 2y &= 8 \end{aligned}$$	$$\begin{aligned} x + y &= 4 \\ x + y &= 3 \end{aligned}$$

The Method of Elimination

The Method of Elimination

Solving a system with the method of elimination:

This method is based on the fact that if $a = b$ and $c = d$, then $a + c = b + d$ and $a - c = b - d$.

- Choose one variable to eliminate, say x
- If necessary, multiply multiple one or both equations so that equation has the same coefficient of x but opposite sign
- Add the two equations, eliminating x, solve to obtain a value for y
- Plug this value for y back into either of the equations to obtain a value for x

Examples:

$$\begin{aligned} x + y &= 4 \\ 2x - y &= 3 \end{aligned} \qquad \begin{aligned} x + y &= 4 \\ 2x + 2y &= 8 \end{aligned} \qquad \begin{aligned} x + y &= 4 \\ x + y &= 3 \end{aligned}$$

Systems of Linear Equations in Three Variables

Systems of Equations in Three Variables

We can have systems of three equations with three variables

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

Behaves the same as systems of two equations:

- Can be solved via substitution or elimination
- Can have 0, 1, or infinite solutions

Solving with Elimination and Substitution

Solve the systems via substitution and elimination:

One thousand tickets were sold for some play, generating \$3800 dollars in revenue. The ticket prices were \$3 for children, \$4 for students, and \$5 for adults. Also, there were 100 more adult tickets sold than student tickets.

Let C, S, A be the prices of child, student, and adult tickets, respectively. Derive and solve a system of equations relating the tickets sales for these variables.

$$
\begin{aligned}
C + S + A &= 1000 \\
3C + 4S + 5A &= 3800 \\
A &= S + 100
\end{aligned}
$$

Solving with Elimination and Substitution

Solve the systems via substitution and elimination:

At a restaurant, three people order some food...

Person 1 buys 2 burger, 1 order of fries, and 1 soda, paying \$9.
Person 2 buys 1 burger, 2 order of fries, and 1 soda, paying \$8.
Person 3 buys 3 burger, 3 order of fries, and 2 soda, paying \$18

Let B, F, S be the prices of a burger, fries, and soda, respectively. Derive and solve a system of equations relating the prices and orders of the people for these variables. In other words. find what prices the foods need to be in order to achieve the bills given for each person.

Solving with Elimination and Substitution

Solve the systems via substitution and elimination:

Person 1 buys 2 burger, 1 order of fries, and 1 soda, paying \$9.
Person 2 buys 1 burger, 2 order of fries, and 1 soda, paying \$8.
Person 3 buys 3 burger, 3 order of fries, and 2 soda, paying \$18

$$2B + F + S = 9$$
$$B + 2F + S = 8$$
$$3B + 3F + 2S = 18$$

Solving with Elimination and Substitution

Solve the systems via substitution and elimination:

$$
\begin{aligned}
x + y - z &= -2 \\
x + 2y - 2z &= -3 \\
y - z &= -1
\end{aligned}
$$

Answer: $(-1, z - 1, z)$

Representing Systems of Linear Equations with Matrices

Representing Systems of Linear Equations with Matrices

What are matrices (singular: matrix):

Definitions:

- **Matrix**: an array of numbers arranged in a rectangular fashion
- **Entry/element**: the numbers in the matrix
- **Dimension**: the number of rows and columns in a matrix $(m \times n)$
- **Square matrix**: a matrix with equal numbers of rows and columns $(n \times n)$

$$\underset{2 \times 2}{\begin{bmatrix} 4 & -7 \\ -2 & 9 \end{bmatrix}} \quad \underset{3 \times 3}{\begin{bmatrix} -1 & -5 & 3 \\ 1.2 & 0 & -1.3 \\ 4.1 & 5 & 7 \end{bmatrix}} \quad \underset{4 \times 4}{\begin{bmatrix} -3 & -6 & 9 & 5 \\ \sqrt{2} & -8 & -8 & 0 \\ 3 & 0 & 19 & -7 \\ -11 & -3 & 7 & 8 \end{bmatrix}} \quad \underset{3 \times 2}{\begin{bmatrix} 5 & -2 \\ -2 & \pi \\ 1 & -1 \end{bmatrix}} \quad \underset{2 \times 3}{\begin{bmatrix} 1 & -0.5 & 9 \\ 5 & 0.4 & -3 \end{bmatrix}}$$

Figure: Source: our textbook

Representing Systems of Linear Equations with Matrices

See the Concept: Representing a Linear System with a Matrix

A	B	C
System of Three Equations	**Coefficient Matrix**	**Augmented Matrix**

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \qquad \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \qquad \left[\begin{array}{ccc|c} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{array}\right]$$

- **A** The a_k, b_k, c_k, and d_k are constants and x, y, and z are variables.
- **B** The coefficients of the variables are represented in a square matrix called the **coefficient matrix** of the linear system.
- **C** The matrix is enlarged to include the constants d_k. The vertical line in this matrix corresponds to where the equals sign occurs in each equation. This matrix is commonly called an **augmented matrix.**

Figure: Source: our textbook

Representing Systems of Linear Equations with Matrices

Examples of representing a system with a matrix:

$$\begin{aligned} 3x - 4y &= 6 \\ -5x + y &= -5 \end{aligned} \qquad \begin{aligned} 2x - 5y + 6z &= -3 \\ 3x + 7y - 3z &= 8 \\ x + 7y &= 5 \end{aligned}$$

$$\left[\begin{array}{rr|r} 3 & -4 & 6 \\ -5 & 1 & -5 \end{array}\right] \qquad \left[\begin{array}{rrr|r} 2 & -5 & 6 & -3 \\ 3 & 7 & -3 & 8 \\ 1 & 7 & 0 & 5 \end{array}\right]$$

Figure: Source: our textbook

Row-Echelon Form

Row-Echelon Form

What is row-echelon form:

Row-echelon form is way of writing a matrix which makes solving the system that it represents simple and easy

Properties:

- The first non-zero entry in each row of the matrix is 1
- The main diagonal has only 1's and 0's

Examples of row-echelon form (main diagonal highlighted blue):

$$\begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 1 & -6 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 & -1 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$

Figure: Source: our textbook

Row-Echelon Form

Back-substitution using row-echelon form:

Back-substitution is an easy way of solving a linear system written row-echelon form

Steps:

- The bottom row gives us the value of z
- Plug z into the equation represented by the row above to get y
- Plug y, z into the equation represented by the row above to get x

$$\left[\begin{array}{rrr|r} 1 & 1 & 3 & 12 \\ 0 & 1 & -2 & -4 \\ 0 & 0 & 1 & 3 \end{array}\right] \qquad \left[\begin{array}{rrr|r} 1 & -1 & 5 & 5 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Figure: Source: our textbook

Gaussian Elimination

Gaussian Elimination

Obviously, not every system/matrix is given to us already in row-echelon form; we must find it ourselves

Gaussian elimination:

Gaussian elimination is the technique for taking a system/matrix which is NOT in row-echelon form and converting it into row-echelon form

This is done by applying **row transformations** to the matrix which are identical to performing the elimination method on the system of equations

Allowed transformation:

- Any two rows can be interchanged
- Entries on any row can be multiplied by a non-zero constant
- Any row may be altered by adding/subtracting a multiple of another row to/from it

Gaussian Elimination

Gaussian elimination:

Allowed transformation:

- Any two rows can be interchanged
- Entries on any row can be multiplied by a non-zero constant
- Any row may be altered by adding/subtracting a multiple of another row to/from it

Here is an example

Linear System	***Augmented Matrix***
$\begin{aligned} 2x + 4y + 4z &= 4 \\ x + 3y + z &= 4 \\ -x + 3y + 2z &= -1 \end{aligned}$	$\left[\begin{array}{ccc\|c} 2 & 4 & 4 & 4 \\ 1 & 3 & 1 & 4 \\ -1 & 3 & 2 & -1 \end{array}\right]$

Figure: Source: our textbook

Row-Echelon Form

Gaussian elimination:

Allowed transformation:

- Any two rows can be interchanged
- Entries on any row can be multiplied by a non-zero constant
- Any row may be altered by adding/subtracting a multiple of another row to/from it

Here is an example with NO solutions (also find the augmented matrix)

$$\begin{aligned} x - 2y + 3z &= 2 \\ 2x + 3y + 2z &= 7 \\ 4x - y + 8z &= 8 \end{aligned}$$

Figure: Source: our textbook